

24AD6501 DEEP LEARNING L T P C 3 0 0 3 OBJECTIVES: • To understand and need and principles of deep neural networks. • To understand CNN and RNN architectures of deep neural networks. • To comprehend advanced deep learning models. • To learn the evaluation metrics for deep learning models. • To understand the concepts of autoencoders and generative models. UNIT I DEEP NETWORKS BASICS 9 Probability Distributions - Gradient based Optimization - Machine Learning Basics: Capacity - Overfitting and underfitting - Hyperparameters and validation sets - Estimators - Bias and variance - Stochastic gradient descent - Challenges motivating deep learning - Deep Networks: Deep feedforward networks - Regularization - Optimization. UNIT II CONVOLUTIONAL NEURAL NETWORKS 9 Convolution Operation - Sparse Interactions -- Parameter Sharing - Equivariance - Pooling - Convolution Variants: Strided - Tiled - Transposed and dilated convolutions - CNN Learning: Nonlinearity Functions - Loss Functions - Regularization -- Optimizers - Gradient Computation. UNIT III RECURRENT AND RECURSIVE NETS 9 Unfolding Graphs - Recurrent Neural Networks - Bidirectional RNN - Sequence to Sequence RNN - Deep Recurrent Networks - Recursive Neural Networks - Long Term Dependencies; Leaky Units: Skip connections and dropouts - Gated Architecture: LSTM. UNIT IV MODEL EVALUATION 9 Performance metrics - Baseline Models - Selecting Hyperparameters - Debugging strategies - Applications UNIT V AUTOENCODERS AND GENERATIVE MODELS 9 Autoencoders: Undercomplete autoencoders - Regularized autoencoders - Stochastic encoders and decoders - Learning with autoencoders; Variational autoencoders; Generative adversarial networks.